

Machine-learning score to predict in-hospital outcomes in patients Hospitalized in Intensive Cardiac Care Unit

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Background: Few scores available for risk stratification of patients hospitalized in Intensive Cardiac Care Unit (ICCU) and benefit of machine learning (ML) not established.

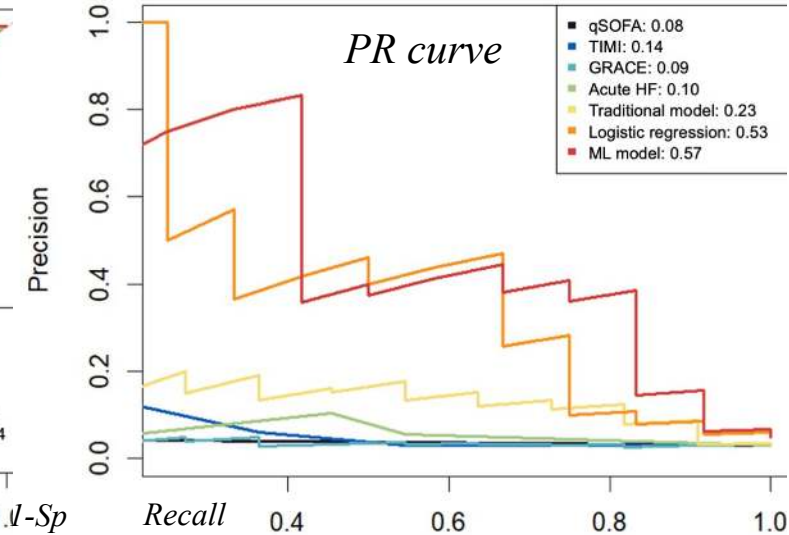
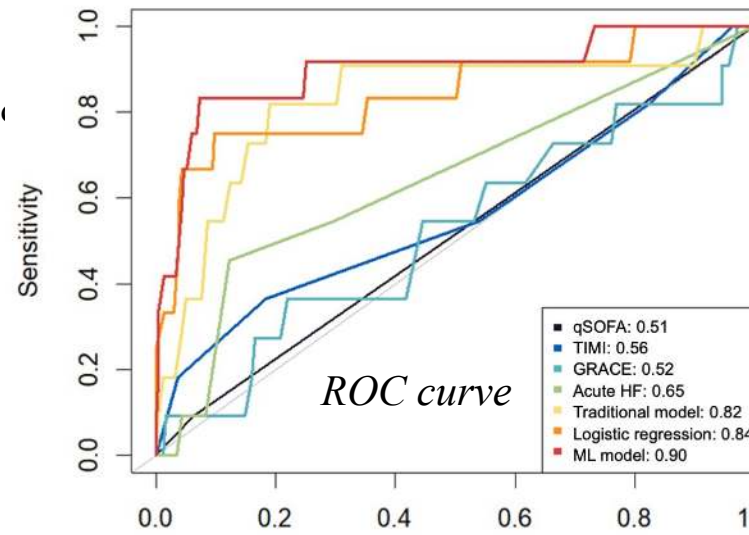
Purpose: Investigate the feasibility and accuracy of ML to predict in-hospital major adverse events (MAE) in patients hospitalized in ICCU and compare its performance with existing scores.

- MAE : composite of all-cause in-hospital mortality, cardiogenic shock and resuscitated cardiac arrest

Methods: In April 2021, a French nationwide, multicenter, prospective, study (39 centers) included all consecutive patients admitted in ICCU (N=1499), primary outcome : in-hospital MAE.

- 28 clinical, biological, ECG, and echocardiographic variables considered for Boruta variable selection algorithm
- 31 random centers = index cohort, 8 other centers = external validation cohort
- Several ML algorithms (logistic regression, ridge regression, boosted cost-sensitive C5.0 and XGBoost. C5.0) assessed to build a ML-score to predict in-hospital MAE

Conclusion: with 5 variables: mean arterial pressure, exhaled CO level, left ventricular ejection fraction, TAPSE value and peak E/e' ratio, ML score exhibited a better prognostic value to predict in-hospital outcomes compared with existing scores in patients admitted in ICCU.



Results: 1499 consecutive patients included (age 63.3 ± 14.9 years, 69.6% male), with 61 in-hospital MAE (4.3%).

- **7 variables** selected in the training set (N=844) : illicit drug use, mean arterial pressure, Killip stage, exhaled CO level, left ventricular ejection fraction, TAPSE value and peak E/e' ratio.
- Boosted cost-sensitive C5.0 technique showed the best performance (AUC=0.90, PR-AUC=0.57, F1 score=0.5), based on 5 of 7 variables : **mean arterial pressure, exhaled CO level, left ventricular ejection fraction, TAPSE value and peak E/e' ratio**. Good AUC in the external cohort (AUC 0.97).
- **Improved performance** compared with existing scores (ML score: 0.90 vs TIMI-score: 0.56, GRACE-score: 0.52, Acute HF-score: 0.65; all $p < 0.001$).